

## 3432. Count Partitions with Even Sum Difference

**Difficulty:** Easy

**Link:** <https://leetcode.com/problems/count-partitions-with-even-sum-difference/description/?envType=daily-question&envId=2025-12-05>

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### Problem:

You are given an integer array `nums` of length `n`.

A **partition** is defined as an index `i` where  $0 \leq i < n - 1$ , splitting the array into two **non-empty** subarrays such that:

- Left subarray contains indices `[0, i]`.
- Right subarray contains indices `[i + 1, n - 1]`.

Return the number of **partitions** where the **difference** between the **sum** of the left and right subarrays is **even**.

### Example 1:

**Input:** `nums = [10,10,3,7,6]`

**Output:** 4

**Explanation:**

The 4 partitions are:

- [10] , [10, 3, 7, 6] with a sum difference of  $10 - 26 = -16$  , which is even.
- [10, 10] , [3, 7, 6] with a sum difference of  $20 - 16 = 4$  , which is even.
- [10, 10, 3] , [7, 6] with a sum difference of  $23 - 13 = 10$  , which is even.
- [10, 10, 3, 7] , [6] with a sum difference of  $30 - 6 = 24$  , which is even.

### Example 2:

**Input:** nums = [1,2,2]

**Output:** 0

**Explanation:**

No partition results in an even sum difference.

### Example 3:

**Input:** nums = [2,4,6,8]

**Output:** 3

**Explanation:**

All partitions result in an even sum difference.

**Constraints:**

- $2 \leq n == \text{nums.length} \leq 100$

- `1 <= nums[i] <= 100`